LAB CONSOLE — MANUAL

OPERATOR GUIDE · ADMIN GUIDE · TECHNICAL REFERENCE · PRINTED FROM THE TECHNICAL ARCHIVE

Lab Console is a wall-mountable web dashboard for monitoring and controlling a fleet of Bambu Lab 3D printers (1–10) over the local network. It runs on a Raspberry Pi 3 or newer and is viewed in any browser — typically a small touch screen in the lab. The fleet view gives at-a-glance state for every printer; tapping a panel opens a full console with live camera, telemetry, and controls including pause, resume, stop, and starting new prints.

00 · ARCHITECTURE AT A GLANCE

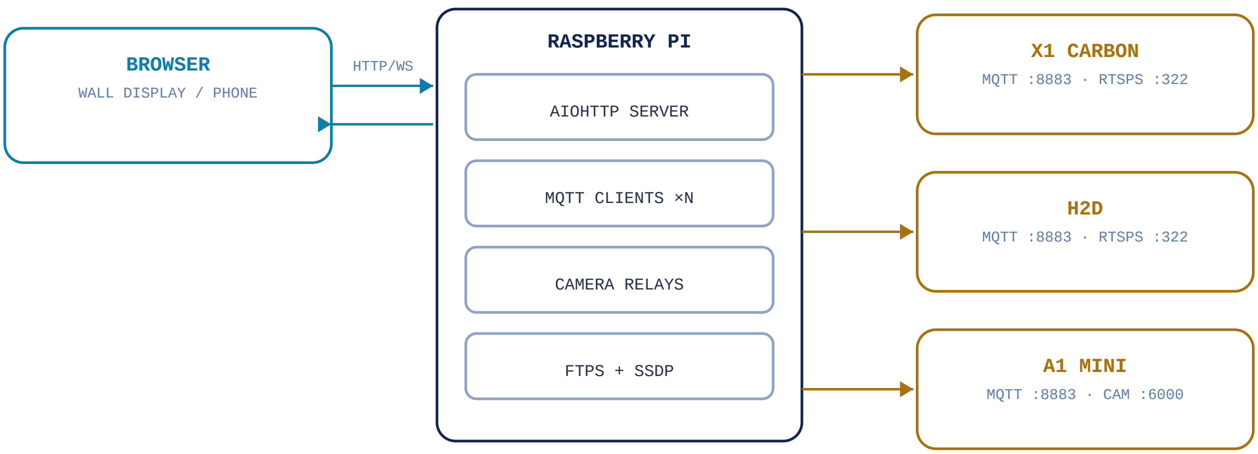

FIG 0.1 — ONE MQTT LINK PER PRINTER; CAMERAS RELAYED ONE AT A TIME; STATE PUSHED TO BROWSERS OVER A WEBSOCKET.

The Pi holds one always-on MQTT connection per printer (Developer Mode, TLS, port 8883) for status and control, relays one live camera at a time as MJPEG, moves print files over FTPS, and listens for SSDP announcements so printers are found again after DHCP changes.

> ■ **DEMO MODE**
> Run python3 server.py --demo for a fully simulated fleet — every state on a timer, working controls, no printers contacted. A violet DEMO badge shows in the status bar. Ideal for training.

OPERATOR GUIDE

READING AND OPERATING THE FLEET CONSOLE

01 · READING THE DASHBOARD

The status bar shows fleet totals, the clock, and a **LINK** indicator for the browser's connection to the Pi — if LINK is red, the view may be stale. Each printer is a bezel-framed panel that cycles every ~8 seconds between its status readout, an ambient scanner animation (idle printers only), and — for one printer at a time — the live camera slot. The state lamp in the panel header never cycles away. Tap any panel for its detail console.

02 · PRINTER STATES

IDLE PRINTING PAUSED ERROR OFFLINE

IDLE	Steady cyan lamp. Online and ready; readout shows temperatures and the last job.
PRINTING	Pulsing amber. File name, layer count, temperatures, time remaining, progress.
PAUSED	Steady violet. Job held; resume from the detail view.
ERROR	Blinking red lamp **and the entire panel bezel flashes red in sync**. Readout lists HMS code(s). Error panels never cycle away.
OFFLINE	Dark lamp. No MQTT link — check power, network, Developer Mode.

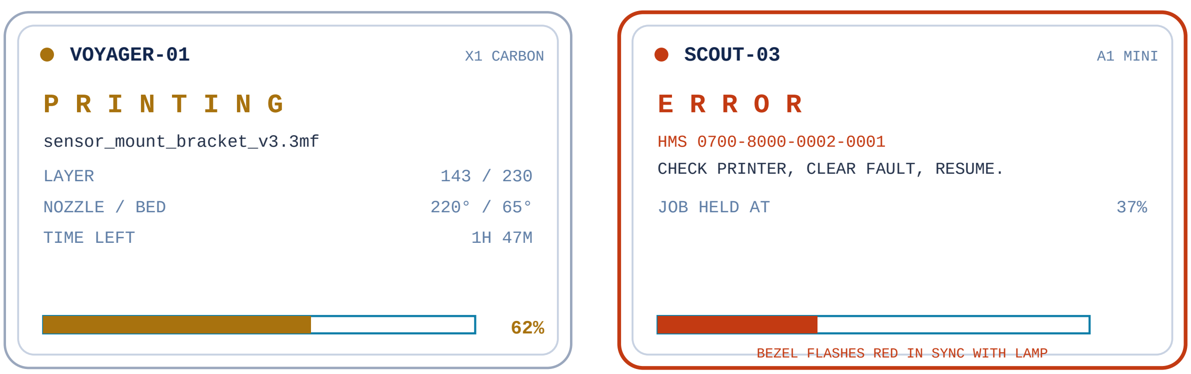

FIG 2.1 — A PRINTING PANEL AND AN ERROR PANEL AS DRAWN ON THE CONSOLE.

03 · CAMERA VIEWS

One camera relay runs at a time; the dashboard's live slot rotates through the fleet, and opening a detail view pins that printer's camera. X1 Carbon / H2D provide real video (RTSPS, relayed at reduced frame rate); A1 / A1 Mini provide roughly one frame per second from the chamber camera — the slideshow feel is the hardware's design, not a fault.

> ■ **NOTE**
> The white ● LIVE tag is an indicator, not an alarm — red is reserved exclusively for error signaling. A grey NO SIGNAL tag means liveview is unavailable; status and controls still work.

04 · CONTROLS

CONTROL	BEHAVIOR
■ **PAUSE / ■ RESUME**	Holds or resumes the job; the button swaps by state. Resume also clears a recoverable ERROR (e.g. after feeding filament).
■ **STOP**	Aborts the job. Always asks for confirmation first.
■ **LIGHT**	Toggles the chamber light — useful before judging a camera view.
▲ **START PRINT…**	Pick a sliced .3mf on the printer’s SD card, or upload one. Selecting a file arms START.

> ▲ **CAUTION**
> STOP cannot be undone — a stopped job restarts from zero. The console always asks for confirmation before sending it.

> ■ **GOOD PRACTICE**
> Before starting a print remotely, use the camera to confirm the bed is clear.

ADMIN GUIDE

SETUP, CONFIGURATION, AND CARE OF THE CONSOLE

01 · PRINTER PREREQUISITES (EACH PRINTER)

- Update to recent firmware.
- Switch to **LAN Only Mode** and enable **Developer Mode** (Settings → General/Network). This opens the local MQTT, live-stream, and FTP channels. Developer Mode printers cannot simultaneously connect to Bambu Cloud.
- Record the **Access Code** (Settings → Network) and **Serial Number** (Settings → Device).
- **X1-series / H2D only:** enable **LAN Mode Liveview** for the camera.

> **NOTE**
> Regenerating a printer's access code invalidates the one in config.json — the panel goes OFFLINE until the config is updated.

02 · INSTALLATION

```
sudo apt update && sudo apt install -y python3-pip ffmpeg
cd /home/pi/lab-console
pip3 install -r requirements.txt --break-system-packages
cp config.example.json config.json   # fill in serials + access codes
python3 server.py                    # open http://:8080
```

RUNNING ON UBUNTU / OTHER LINUX HOSTS

The console runs unchanged on a standard Ubuntu machine (desktop or server) instead of a Raspberry Pi — nothing in the code is Pi-specific. Install the same way, shown above. A few environment differences to check:

- **Firewall** — Ubuntu often runs ufw, which Raspberry Pi OS does not enable by default. Allow the port: sudo ufw allow 8080/tcp.
- **Same LAN as the printers** — SSDP discovery is passive; the host must sit on the same network segment as the printers, not behind a router hop or separate VLAN.
- **Laptops** — Wi-Fi power-saving can drop the idle discovery socket or let the machine sleep. Disable suspend/Wi-Fi power management, or use a wired/always-on machine.

> **NOTE**
> A typical Ubuntu machine has far more headroom than a Pi 3 — camera.max_streams in the config can safely be raised above 1 for multiple simultaneous live camera views.

03 · CONFIG REFERENCE — PER PRINTER

FIELD	MEANING
id	Short unique key; used in URLs.

FIELD	MEANING
name / model	Display name and model label on the dashboard.
serial	Printer serial — used for MQTT topics and SSDP matching.
access_code	LAN access code from the printer screen.
ip	Optional; discovery fills and updates it. A static/reserved address makes startup instant.
camera	rtsp (X1-series, H2D) · chamber (A1/A1 Mini/P1) · none.

CONFIG REFERENCE — GLOBAL

FIELD	DEFAULT	MEANING
http_port	8080	Port the console serves on.
max_upload_mb	200	Upload size limit.
camera.max_streams	1	Concurrent camera relays; keep 1 on a Pi 3. New viewers evict the oldest.
camera.rtsp_fps / rtsp_width	8 / 640	RTSPS transcode cost knobs.
sdcard_url_base	file:///sdcard/	Switch to file:///mnt/sdcard/ if start-print reports file-not-found.

04 · RUN AT BOOT / KIOSK DISPLAY

```
sudo cp lab-console.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now lab-console
journalctl -u lab-console -f     # logs
```

For a wall display, run a kiosk browser: chromium-browser --kiosk http://localhost:8080. The frontend pauses panel cycling while the tab is hidden.

05 · TROUBLESHOOTING

SYMPTOM	LIKELY CAUSE / FIX
Panel OFFLINE	Printer unreachable over MQTT. Check power, LAN, Developer Mode, access code. Discovery re-finds moved printers within about a minute.
X1/H2D camera NO SIGNAL	LAN Mode Liveview off, or ffmpeg missing. Some H2D firmware temporarily shipped without local liveview — status/control keep working.
A1 camera ~1 fps	Hardware rate of the chamber camera; not a fault.

SYMPTOM	LIKELY CAUSE / FIX
Start print: file not found	Switch sdcard_url_base to file:///mnt/sdcard/.
SD list empty	Only SD root files are listed; uploads go to the root.
LINK red in status bar	Browser lost its WebSocket; it reconnects automatically. If persistent, check the service logs.

▲ SECURITY
No login; access codes live in config.json. Keep the console on a trusted LAN/VLAN and do not expose port 8080 to the internet.

TECHNICAL REFERENCE

PROTOCOLS, SCHEMAS, AND THE HTTP API

Printer-side protocols are the community-documented LAN Developer Mode interfaces (OpenBambuAPI; Bambu Lab wiki, third-party integration). Nothing uses Bambu Cloud.

01 · MQTT (STATUS + CONTROL)

One TLS connection per printer to port **8883**, username bblp, password = access code, self-signed certificate accepted.

```
device//report    <-  status pushes (partial JSON, merged)
device//request   ->  commands
```

ACTION	PAYLOAD (ABBREVIATED)
full refresh	{"pushing":{"command":"pushall"}}
pause / resume / stop	{"print":{"command":"pause"|"resume"|"stop","param":""}}
chamber light	{"system":{"command":"ledctrl","led_node":"chamber_light","led_mode":"on"|"off",…}}
start print	{"print":{"command":"project_file","url":"file:///sdcard/.3mf","param":"Metadata/plate_1.gcode","use_ams":true,…}}

Report fields read: gcode_state (RUNNING/PREPARE/PAUSE/IDLE/FINISH/FAILED), mc_percent, mc_remaining_time, layer_num/total_layer_num, subtask_name/gcode_file, temperatures and targets, cooling_fan_speed, spd_lvl_name, lights_report, AMS tray types/colors, and hms / print_error for the ERROR state.

02 · CAMERA PROTOCOLS

MODELS	TRANSPORT	CONSOLE RELAY
X1 / X1C / X1E / H2D	RTSPS :322 — rtsps://bblp:@:322/streaming/live/1. Requires LAN Mode Liveview.	ffmpeg → MJPEG at configurable fps/width; served as multipart/x-mixed-replace.
A1 / A1 Mini / P1	Proprietary chamber-image service, TLS :6000. 80-byte auth packet, then length-prefixed JPEG frames at ~1 fps.	Frames pass straight through — no transcoding.

The relay hub enforces camera.max_streams (default 1); a new viewer evicts the oldest. The frame queue is depth-2 and drops stale frames so slow clients see the latest image rather than building latency.

A chamber camera needs roughly three seconds to hand over its first frame, so the hub keeps the newest frame per printer and replays it the instant a viewer connects, then overwrites it from the live stream. The replay is skipped once that frame is more than 30 seconds old, so the first image is never badly stale.

03 · FTPS + DISCOVERY

FTPS — implicit TLS on port 990, user bblp. Lists .3mf/.gcode in the SD root (MLSD with NLST fallback) and uploads new files ahead of a project_file command.

SSDP — passive listener on UDP 2021 and 1990 for printer announcements, matching serial in USN and address in Location. On address change the MQTT client reconnects. If another app holds the port, discovery is skipped and configured IPs are used.

04 · WEBSOCKET SCHEMA

```
{ "type":"fleet", "version":123, "demo":false, "printers":[ {
    "id":"voyager01","name":"VOYAGER-01","model":"X1 CARBON",
    "state":"printing",             // idle|printing|paused|error|offline
    "online":true,"camera":"rtsp",  // rtsp|chamber|demo|none
    "file":"bracket_v3.3mf","percent":62,"remaining_min":107,
    "layer":143,"total_layers":230,"nozzle":220,"nozzle_target":220,
    "bed":65,"bed_target":65,"chamber":41,"fan":100,
    "speed":"STANDARD","light":"on",
    "ams":[{"type":"PLA","color":"#3fd8ff"},...],
    "hms":[{"attr":...,"code":...}],"print_error":0 } ] }
```

Client → server, acknowledged with {"type":"ack"} or {"type":"error"}:

```
{ "type":"cmd", "id":"voyager01",
  "action":"pause|resume|stop|light|start_print|refresh",
  "on":true,          // light only
  "file":"part.3mf" } // start_print only
```

05 · HTTP API

ENDPOINT	METHOD	PURPOSE
/	GET	The console UI.
/ws	GET	WebSocket (schema above).
/cam/	GET	MJPEG stream for one printer.
/api//files	GET	JSON list of printable files on the SD card.
/api//upload	POST	Multipart upload of a .3mf/.gcode to the SD card.
/docs	GET	The Technical Archive.